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Perspective

# The inheritance of mitochondrial DNA heteroplasmy:

## random drift, selection or both?

**The mammalian mitochondrial genome (mtDNA) is a small double-stranded DNA molecule that is exclusively transmitted down the maternal line. Pathogenic mtDNA mutations are usually heteroplasmic, with a mixture of mutant and wild-type mtDNA within the same organism. A woman harbouring one of these mutations transmits a variable amount of mutant mtDNA to each offspring. This can result in a healthy child or an infant with a devastating and fatal neurological disorder. Understanding the biological basis of this uncertainty is one of the principal challenges facing scientists and clinicians in the field of mitochondrial genetics.**

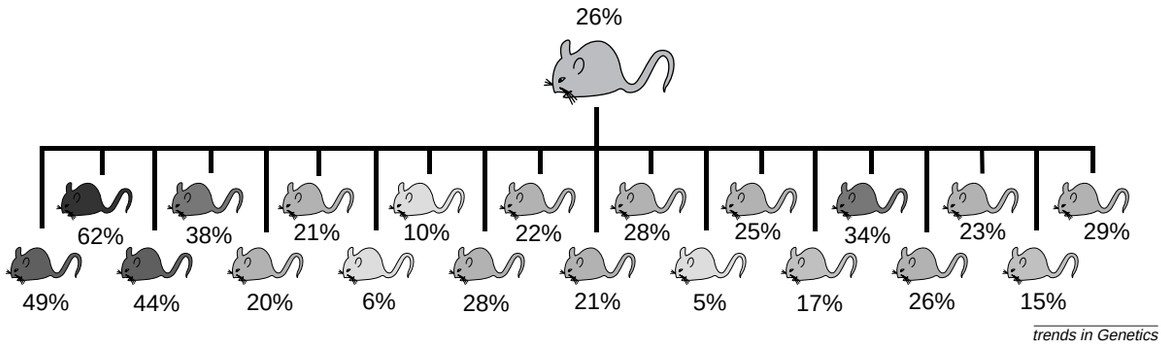
Here we consider the mechanisms behind the transmission of mtDNA heteroplasmy and focus on the key question: is there a selection bias operating during the transmission of heteroplasmic pathogenic mtDNA mutations, or are they inherited like neutral sequence variants?

In addition to the nuclear genome, mammalian cells also have multiple copies of a small (15–17 kb) circle of double-stranded DNA compartmentalized within individual mitochondria[1]. The mammalian mitochondrial genome (mtDNA) codes for 13 polypeptides, each of which is an essential subunit of the mitochondrial respiratory chain. Interspaced among the polypeptide-coding genes are 22 tRNA and two rRNA genes whose products are essential for intramitochondrial protein synthesis[2]. Mammalian mtDNA is compact, with only ~1 kb of noncoding sequence (the D-loop). Unlike nuclear DNA, mtDNA is not coated by protective histones, and it is tethered to the inner mitochondrial membrane, close to the respiratory chain, which is a potent source of oxygen free radicals[3]. These factors are believed to contribute to a



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**FIGURE 1. The transmission of heteroplasmic mtDNA polymorphisms in mice**



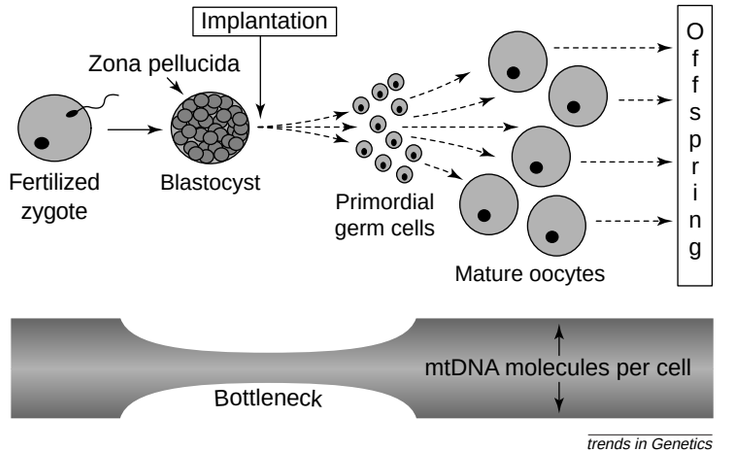

Random genetic drift in a mouse pedigree transmitting heteroplasmic mitochondrial DNA (mtDNA) polymorphisms. Heteroplasmic mice were generated by karyoplast transfer from a NZB donor to a C57Bl6 recipient (see text). The pedigree shown here includes all the offspring of an F1 female mated with a C57Bl6 male. The percentage level of NZB is indicated below a single mother and her offspring. The varying grey scale illustrates the relative proportion of NZB genotype in each mouse.

high rate of mtDNA mutations. The consensus view is that mammalian mtDNA is exclusively inherited down the maternal line. As a result, all the offspring of a single female are clonal for mtDNA. The high mutation rate and clonal maternal transmission lead to the accumulation of novel mtDNA mutations within the population.

In the 1980s, genetic studies of Holstein cows revealed mtDNA single-nucleotide polymorphisms (SNPs) in different members of the same pedigree. Only one allele was detected in the blood of the founder female, but many of her offspring had varied levels of mtDNA heteroplasmy, and the new SNP rapidly became fixed in subsequent generations[4]. In one extreme case, a new mutation occurred and was fixed within a single transmission[5]. One explanation for these rapid changes in genotype is a mitochondrial genetic 'bottleneck', with only a small proportion of the available mitochondrial genomes repopulating the offspring of the next generation. Unequal partitioning of genomes during cytokinesis (vegetative segregation) can lead to differences between daughter cells through random genetic drift. This 'sampling' has minimal effect when the cells involved contain large numbers (~50 000 or more) of freely intermixing mtDNA molecules. However, if the effective number of mtDNA molecules is reduced (a 'bottleneck'), then the rate and size of genotypic shifts increase dramatically. In addition to a simple reduction in the amount of mtDNA, this could arise through the selective replication of only a small group of mtDNA molecules, which eventually repopulate the next generation[6].

At first sight, it seems counterintuitive that such a mechanism should have evolved, but the process must be seen in a wider context. Thirty-five years ago, Müller showed that asexual organisms with high mutation rates will accumulate mildly deleterious mutations over generations[7]. In time, these mutations will compromise the function of individual cells, then the whole organism, and – eventually – the species. Mammalian mitochondrial genes should be susceptible to Müller's 'ratchet', but the mitochondrial genetic bottleneck might have evolved to counteract this process[8]. A germ-line bottleneck results in rapid genetic drift. This would lead to the direct loss of the mutation through drift, or the formation of individuals that are homoplasmic for the new mutant mtDNA species,

**FIGURE 2. The mitochondrial genetic bottleneck**

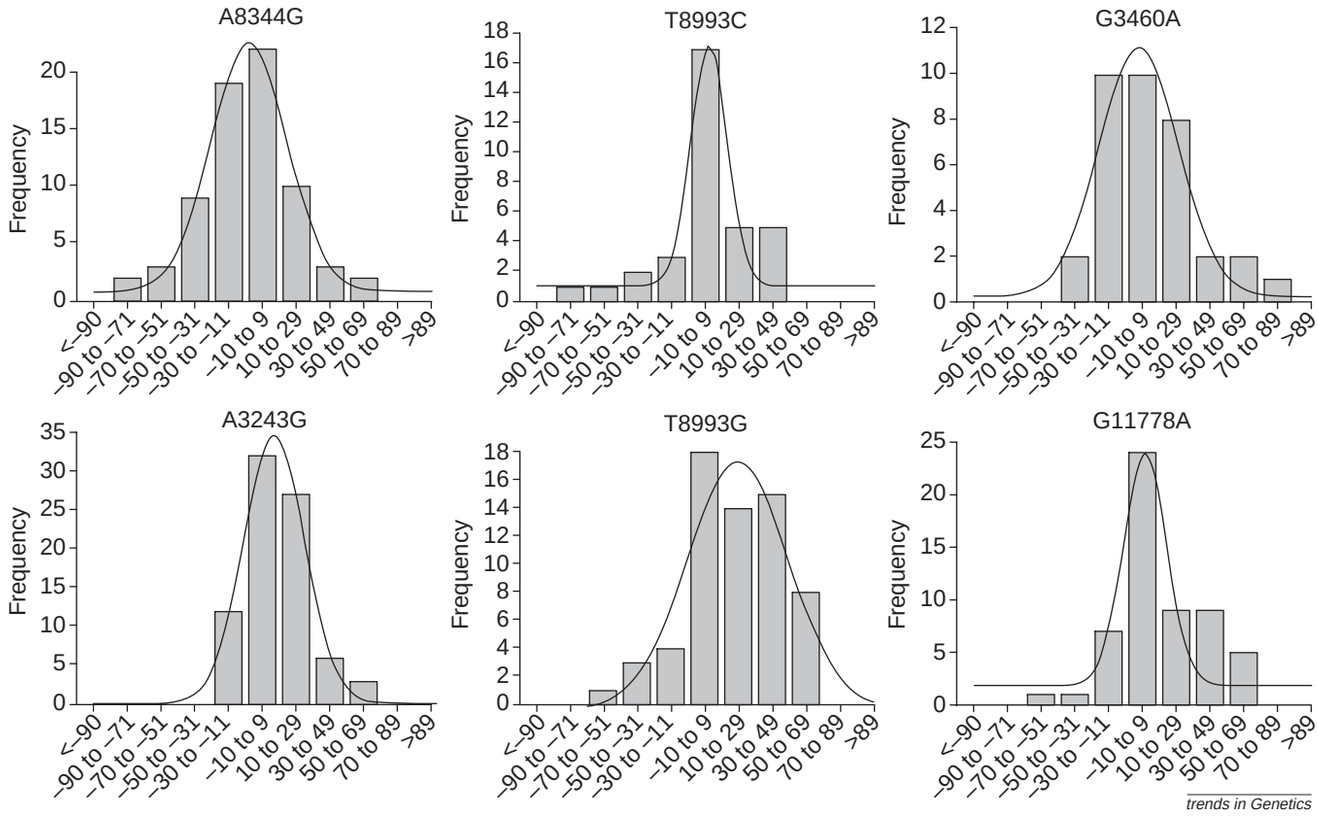



Preimplantation embryonic development and the formation of the female germ line. Diagram showing a reduction in the number of mitochondrial DNA (mtDNA) molecules per cell – the mitochondrial genetic 'bottleneck'.

swiftly exposing the new mutation to the forces of natural selection operating at the level of the individual organism. A germ-line bottleneck consequently leads to the loss of mildly deleterious mutations before they can accumulate within the population. In the short term, however, the bottleneck can also cause major problems for the individual, because those mutations that 'slip' through will tend to become homoplasmic much more rapidly.

Similar dramatic changes in the level of heteroplasmy have also been seen in human pedigrees, including mutations that appear to become fixed within one generation[9,10]. In humans, mtDNA heteroplasmy is often associated with disease[11,12]. For the more common pathogenic mutations, a high percentage of mutant mtDNA in postmitotic tissues (such as skeletal muscle) correlates with the severity of the neuromuscular disorder[13,14]. Consequently, rapid shifts in the level of heteroplasmy within a pedigree are associated with a range of different clinical phenotypes

**FIGURE 3. The transmission of pathogenic heteroplasmic mtDNA mutations in humans**




Histogram of frequency distribution of $O - M$ (the difference in mutant load between offspring and their corresponding mothers) for pathogenic mitochondrial DNA (mtDNA) point mutations in humans. Pedigrees were studied either in the laboratory of the authors, or obtained from the literature as described previously[14,34]. For each transmission, we calculated $O - M$, the difference between the percentage of mutated mtDNA in the offspring ($O$) and the percentage of mutated mtDNA in the mother ($M$). Values of $O - M$ that included a proband were omitted to minimize ascertainment bias. Blood values were used for the A8344G, G3460A, A3243G and G11778A mutations. A variety of different tissues were analysed for the T8993G/C mutations, but analyses have shown that for these mutations, the level of heteroplasmy does not vary between tissues or over time[10]. Values of $O - M$ were assigned to one of ten categories (see the *x*-axis of the histograms for the category ranges), and the frequency distributions for these values were plotted. From first principles, the distribution of $O - M$ should be binomial. However, because we identified a large number of transmissions, the distribution closely approximates a gaussian curve. This was supported by Kolmogorov–Smirnov analysis (see Table 1). Gaussian curves were fitted to the histogram using Origin v6 (Microcal Software), and descriptive gaussian statistics were calculated. We tested whether the mean $O - M$ value was statistically significantly different from zero using a one sample *t*-test [Minitab v12 (Minitab Inc.)]. It is likely that the width of the distributions is determined by the size of the genetic bottleneck (the effective number of mtDNA transmission units), and it is interesting that the standard deviations of the six distributions are of the same order of magnitude, but a more accurate comparison will only be possible with much larger numbers of independent transmissions.

– from asymptomatic individuals to severely affected offspring with a progressive fatal neuromuscular disease. An understanding of this process thus has implications for the origin and fixation of mtDNA mutations during evolution[15], as well as the clinical management and counselling of patients with mtDNA diseases[16].

## The inheritance of heteroplasmic polymorphisms in mice

To advance our understanding of the mechanisms of transmission of mtDNA heteroplasmy, a number of groups have developed heteroplasmic mice by karyoplast (cytoplast) transfer (White *et al.*, unpublished; Refs 17–20). Extensive studies have been carried out on large pedigrees of these mice that transmit heteroplasmic and apparently neutral mtDNA polymorphisms. The offspring of a single heteroplasmic mouse can have dramatically different levels of heteroplasmy, but the mean level of heteroplasmy among all the offspring is approximately equal to the proportion of mutated mtDNA in the mother[17] (Fig. 1). Because the genotypes of the progeny are distributed evenly above and below the mean level, these experiments indicate that, for polymorphisms, the transmission of mtDNA heteroplasmy to the offspring is predominantly determined by random genetic drift[17]. Studies of single cells from the developing germ line of embryonic heteroplasmic mice suggest that the restriction in mtDNA copy number occurs at a very early stage during the *in utero* development of a female mouse destined to become the mother of the subsequent generation[17] (Fig. 2). The observations in heteroplasmic mice are compelling, but there are also important biological differences between mice and humans[21] that might have a profound effect on the transmission of mtDNA heteroplasmy.

## Differences between mice and humans

Mature human and mouse oocytes contain approximately 100 000 mtDNA molecules[22,23]. After fertilization of the mouse oocyte, there is no measurable mtDNA turnover until the embryo implants, so that the amount of mtDNA

**TABLE 1. Statistical analyses carried out on the *O – M* distributions[a]**

| MtDNA mutation | *n*[b] | Kolmogorov–Smirnov *P*-value[c] | Mean *O – M* (SD)[d] | *O – M* SEM *P*-value[e] | Skewness (SD)[f] |
|---|---|---|---|---|---|
| A8344G | 70 | >0.15 | −7.44 (27.77) | 3.32<0.05 | −0.10 (0.29) |
| A3243G | 80 | >0.15 | 9.64 (18.98) | 2.12<0.01 | 0.36 (0.27) |
| T8993C | 34 | 0.105 | 0.88 (27.39) | 4.70 n.s. | −0.54 (0.42) |
| T8993G | 63 | >0.15 | 18.87 (27.17) | 3.42<0.01 | −0.40 (0.30) |
| G11778A | 56 | >0.15 | 12.75 (25.79) | 3.45<0.01 | 0.07 (0.33) |
| G3460A | 35 | >0.15 | 9.09 (30.02) | 5.08 n.s. | 0.91 (0.41) |

[a]Statistics calculated using Excel 2000 (Microsoft), Origin v6 (Microcal Software) and Minitab v12 (Minitab Inc.) using *O – M* distributions shown in Fig. 3.
[b]*n*, number of transmissions after the omission of the probands.
[c]Kolmogorov–Smirnov test of normality. The null hypothesis for this test assumes that the data follow a normal distribution. If the *P*-value of the test is less than 0.05, the null hypothesis is rejected and it is concluded that the data do not follow a normal distribution.
[d]*O – M*, percentage level of mutated mtDNA in the offspring minus the percentage level of mutation in the mother (see legend to Fig. 3 for ascertainment details).
[e]*P*-value from a *z*-test against zero using the Students *t*-distribution.
[f]The asymmetry of a distribution about its mean is measured by the skewness of the distribution. The SD on the skewness was calculated by sqrt(6/n).
Abbreviations: SD, standard deviation; *O – M* SEM, standard error of the mean *O – M* value; n.s., not significant.

in the embryo remains constant until the blastocyst stage[24]. During this period, the embryo (and mtDNA) is constrained within the zona pellucida, where it is compartmentalized into a large number of cells (Fig. 2). Following implantation, there is massive cellular proliferation, and, at some point, mtDNA replication must resume. The timing of this resumption is critical, because it will determine the minimum mtDNA copy number per cell. Any delay, however minor, will result in very low levels of mtDNA within individual cells. In the female mouse 26 000 oocytes develop in the 13.5 days between conception and birth[25], whereas in human females, 6.5 million oocytes must develop over five months postconception[26]. Even with this fairly superficial consideration of the differences between mice and humans, it can be seen that very subtle differences in any of these processes could have a profound effect on the final mechanism of transmission of heteroplasmy. Finally, we do not yet have a mouse model that harbours a pathogenic heteroplasmic mtDNA mutation. For the reasons outlined in the next section, pathogenic mutations might be transmitted quite differently from neutral mutations.

## Selection for and against mutant mtDNA occurs in humans

For some mtDNA mutations, particularly those with severe phenotypic effects, selection might operate during the life span of the individual. In some tissues and organs, selection may lead to a decrease in the amount of mutant mtDNA, whereas the level of mutant mtDNA can increase in others. In rapidly dividing stem cell populations, mutant mtDNA can be removed by selection operating at the level of the cell. Vegetative segregation could lead to high levels of mutant mtDNA in some stem cells, and low levels in others. The cells that contain high mutant loads might fail to divide, but those with a low mutant load will divide normally. In keeping with this simple model, the level of the pathogenic A3243G mutation in blood decreases over time (negative selection)[27]. In contrast, the level of mutant mtDNA can increase in some tissues throughout life, and in a number of cases, the increased mutation load has been associated with the progression of disease[28,29]. At present we do not know how this increase occurs, but it is possible that 'defective mitochondria', containing a high mutation load, selectively proliferate (perhaps as some sort of compensatory process) in response to a respiratory chain defect[30]. In time, this mechanism will lead to an increase in the amount of mutant mtDNA (positive selection) within a postmitotic tissue such as skeletal muscle[31]. The question is, do similar processes of selection operate in the female germ line?

## The inheritance of mtDNA heteroplasmy in humans

It is generally assumed that mtDNA deletions are not transmitted from mother to offspring, since the vast majority of patients who have a single mtDNA deletion do not have any relevant family history. There have been a handful of reports describing the transmission of mtDNA rearrangements, but it has often been unclear whether the rearranged molecules were actually duplications or deletions. We therefore restricted our analysis to human pedigrees transmitting pathogenic mtDNA point mutations. Our analysis largely involved measurements made using blood, which is usually the only tissue that is readily available (see legend to Fig. 3 for details). Data from many transmissions can be simultaneously analysed by generating a frequency distribution for the difference in mutant load between offspring and their corresponding mothers ($O - M$, see legend to Fig. 3). If the transmission of mutations were determined by random drift alone, then one would expect to see a symmetrical bell-shaped curve about a mean of zero. If, however, there were a significant selective advantage for either mutated or wild-type genomes during transmission from one generation to the next, then one would expect to see a shift or skew in this distribution. To minimize ascertainment bias, the clinically affected index cases (probands) were excluded from the analysis.

We studied the transmission of six of the most common pathogenic point mutations: A8344G ($n = 70$); T8993C ($n = 34$); G3460A ($n = 35$); A3243G ($n = 80$); T8993G ($n = 63$); and G11778A ($n = 56$). The results of this analysis are shown in Fig. 3 and Table 1. For the A8344G mutation, the $O - M$ values were symmetrically distributed, but the mean value was significantly less than zero. For two mutations (T8993C and G3460A), the distributions appeared to be skewed in favour of the transmission of mutant genomes, but this did not reach statistical significance. For the remaining three mutations (A3243G, T8993G and G11778A), the mean $O - M$ value was significantly greater than zero, suggesting the preferential transmission of mutant genomes.

All six mutations cause disease through the impairment of mitochondrial oxidative metabolism – so how can we

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explain these apparently contradictory findings? Could it be that transmission is, in fact, determined by random genetic drift, and that the apparent selection suggested by the data is an artefact? First, the differences might simply be a sampling effect, and after accumulating a larger number of transmissions, the distributions might shift towards a mean of zero (regression to the mean). If the two largest data sets are combined (A8344G and A3243G), the $O - M$ histogram is symmetrical about a mean that is not significantly different from zero (mean $O - M = -1.06$, SEM = 2.08).

Second, for some mutations, such as A3243G and G3460A, the level of mutated mtDNA in blood decreases with age[27,32]. This process will produce a positive bias or offset in the $O - M$ distributions for these mutations. A similar process does not affect the A8344G and T8993C mutations, possibly explaining why these mutations do not show a skew in favour of mutant genomes; however, levels of the T8993G mutation also do not change with time[10].

Third, and perhaps the most important, is the effect of ascertainment bias. The analysed pedigrees were identified by the presentation of a clinically affected individual, and families with multiple affected members are more likely to be investigated and published in the literature. Despite efforts to minimize the effects of bias by omission of those probands, this identification process leads to an inherent bias in the analysis of the entire matrilineal pedigree because the proportion of mutated mtDNA has increased to high levels in the offspring of the most recent generations. In large sibships, one will be able to ascertain a larger number of transmissions distant from the clinically affected proband, which should reduce this bias. Analysis of the pedigrees indicates that the average size of a T8993G sibship was small (mean sibship size = 2.6, SD = 1.4), when compared to the A8344G mutation (mean sibship size = 6.8, SD = 5.9). This finding suggests that the T8993G data were probably more heavily influenced by ascertainment bias than the A8344G data.

Another factor to consider is the effect of mtDNA mutations on reproductive fitness, due to either gonadal dysfunction[33], or severe disability and death prior to childbearing age[34]. There are many published examples of children with high mutant loads being born to women with low mutant loads, but few examples of the converse. This 'loss' of transmissions might bias the data towards an artefactual apparent increase in the transmission of mutant genomes.

Finally, the $O - M$ data are also consistent with a weak selection effect operating with each maternal transmission – but in different directions for the different mutations. A tendency for levels of the A8344G mutation to decrease during transmission is what would be expected if there were a mechanism working against deleterious mutant mtDNA. This might operate at the level of the oocyte or at the level of the organism, but how could this occur? The available evidence suggests that high levels of oxidative metabolism are not needed during oogenesis or early embryogenesis. Developing oocytes contain relatively few mitochondria per cell[35], and transgenic mice with a profound deficiency of mtDNA (lacking a critical mtDNA transcription factor, Tfam)[36], or cytochrome *c* (Ref. 37), develop up to or beyond the egg cylinder stage despite severely compromised oxidative phosphorylation. It has been suggested that there is an increased rate of spontaneous abortion in women harbouring the A3243G mutation[38], but this is uncertain, and there is no evidence for an excess of fetal deaths in large series of pedigrees with other mutations (for example, see Ref. 14).

It is even more difficult to explain how the level of the A3243G, T8993G and G11778A mutations might increase with each generation. All of the available evidence suggests that the A3243G mutation decreases with time in rapidly dividing tissues[27], whereas the T8993G and G11778A mutations do not change with time[10]. It is plausible that levels of the A3243G mutation might increase within individual oocytes, suspended in meiosis I for many decades, through the proliferation of mitochondria (as seen in skeletal muscle). But we have studied ovarian tissue from a 45-year-old female with A3243G and we saw no histochemical evidence of mitochondrial proliferation within the oocytes, despite clear evidence of proliferation within the adjacent vascular wall (so-called strongly succinate dehydrogenase-positive vessels[39]). Moreover, mitochondrial proliferation is not a feature recognized in patients with the T8993G and G11778A mutations[40].

As a final point, our observations are relevant to recent work on ovarian follicular atresia. In mammals, a huge proportion of the primary oocytes generated during embryonic development never mature, but involute – a process called ovarian atresia. Atresia continues throughout life, and it is unclear why it should occur. It was recently suggested that ovarian atresia might be a mechanism to slow the accumulation of deleterious mtDNA mutations through the preferential removal of oocytes carrying high mutation loads[41]. If this were the case, then the women harbouring heteroplasmic pathogenic mtDNA mutations should have offspring with significantly less mutant mtDNA. Our observations do not support such a dramatic trend. Similarly, the finding that some mature human oocytes contain near-homoplasmic levels of the T8993G mutation[9], and that about 20% of pedigrees with the T8993G mutation appear to result from *de novo* mutations[14] implies that follicular atresia does not preferentially remove oocytes with high levels of mutant mtDNA.

## Conclusions and future prospects

How can we resolve whether selection or pure random drift is the predominant factor in human mtDNA transmission? So far it has not been possible to generate a female mouse that transmits a heteroplasmic pathogenic mtDNA mutation. This is an area of intense interest[3], and when a murine model for heteroplasmic mtDNA disease does become available, it might help us considerably to address these issues. Unfortunately, there remain important anatomical and physiological differences between mice and humans that might limit the general relevance of such animal models, and to get an absolute answer we will need to continue studies in humans. One possible approach entails single-cell analysis of oocytes from women with mtDNA mutations. This approach will take some time, because the tissue will only become available when the ovaries are removed for other reasons, and in any case the level of heteroplasmy in oocytes might not reflect the level in the offspring. An alternative strategy would be prospectively to identify heteroplasmic individuals through widespread random sampling within the community. This would be a mammoth task, and based upon our knowledge of the prevalence of pathogenic mtDNA mutations[42], a very large number of independent individuals would need

to be screened to identify enough families to allow a systematic study. A consortium has recently been formed to collect and analyse data on the transmission of pathogenic human mtDNA mutations in order to decrease the bias of only using published pedigrees[43]. Interested parties are welcome to contact any of the authors for further information; see ENMC Workshops 1999 (http://enmc.spc.ox.ac.uk/Gen/works992.htm).

Why is it so important to determine whether selection is occurring during the transmission of heteroplasmy in human pedigrees? If there is positive selection, and a tendency to increase mutation load with subsequent generations, this would lead to the earlier onset of a more severe phenotype (genetic and clinical anticipation). In contrast, a tendency to decrease mutation load (negative selection) might simply lead to the loss of a deleterious mutation from the population without clinical effects. Both of these outcomes would have important implications for the counselling of women who harbour pathogenic mtDNA mutations in their germ line. Over the last decade, thousands of families have been given a diagnosis of mtDNA disease, and women regularly ask about the risks of transmitting a mtDNA defect to their offspring.

The data presented here are important because they suggest that, if there is selection in the germline against these pathogenic mutations, it will have negligible effect over a single generation. We believe that random genetic drift is the predominant mechanism behind the transmission of pathogenic heteroplasmic mtDNA mutations in humans. This generates its own uncertainties, but we can now reassure families that there is no underlying tendency for mtDNA disease to be more severe in the next generation.

## Acknowledgements

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## Letters to the Editor

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